

Multiple Choice

1. **Answer:** C
Options: A) good fitting; B) overfitting; C) underfitting; D) all of the above
Note: Correct option: C - underfitting
2. **Answer:** A
Options: A) PCA; B) Decision Tree; C) Linear Regression; D) Naive Bayesian
Note: Correct option: A - PCA
3. **Answer:** A
Options: A) random crop and horizontal flip; B) random crop and vertical flip; C) posterization; D) dithering
Note: Correct option: A - random crop and horizontal flip
4. **Answer:** A
Options: A) Supervised learning; B) Unsupervised learning; C) Clustering; D) None of the above
Note: Correct option: A - Supervised learning
5. **Answer:** B
Options: A) Predicting the amount of rainfall based on various cues; B) Detecting fraudulent credit card transactions; C) Training a robot to solve a maze; D) All of the above
Note: Correct option: B - Detecting fraudulent credit card transactions

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'False, False'.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'True, True'.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'either (a) or (b)'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The polynomial degree. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Transform data to zero mean. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key